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Event attribution and partisanship shape local discussion of climate change after extreme weather

Hilary Boudet[1]*, Leanne Giordono[2], Chad Zanocco[3], Hannah Satein[4] and Hannah Whitley[5]

Extreme weather events may provide opportunities to raise public awareness and spur action to address climate change. Using concepts from the study of social movements, we conducted a systematic comparative-case analysis of 15 communities that experienced extreme weather events in the United States between 2012 and 2015 to identify under what conditions, and through what mechanisms, the experience of an extreme weather event generates community discussion and collective action linked to climate change. Although collective action related to climate change was rare post-event, we observed community discussion about the event's link to climate change in slightly more than half of the cases, especially in Democratic and/or highly educated communities that experienced events for which attribution to climate change is more certain. Our results suggest that, although a single event may have limited impact on discussion or collective action about climate change, partisanship and an event's attribution to climate change matter.

Extreme weather events often capture national attention. Recently, these events have taken on added significance with increasing scientific certainty that some types of extreme events have become more frequent and severe due to human-induced climate change[1]. Pointing to the immediate, personal and visceral impacts of these events, many social scientists have suggested that such events may shift public debate about climate change[2–7], particularly in the US context where the issue of climate change has become increasingly partisan[8]. Extreme weather events can serve as focusing events[9]—sudden, uncommon, destructive events with concentrated impacts on a specific geographical area[10]— potentially facilitating awareness and ultimately collective action[11].

A growing body of literature has sought to better understand the link between personal experience of an extreme weather event and individual climate change beliefs. Some studies have found little impact, and often stress the role of other factors, such as political ideology, in determining individual interpretations of specific events[12,13]. To the extent that attitudinal changes take place after the event, they have been shown to be small and brief[14–16]. The result is an ongoing debate about the role of experience versus pre-existing beliefs in shaping views about climate change[17–19]. Empirical evidence supports both perspectives. A growing consensus is emerging that socio-demographic and physical factors (for example, local weather) are important in shaping views about climate change[20–23].

Studies that examined connections between extreme weather events and climate change discussion or activism are less prevalent[21,24,25]. In terms of individual attitudes and behaviour, where most research has been focused, several studies have demonstrated short-term increases in post-event information seeking about and attention to climate change[16,26–28]. However, this focus on individual attitudes and behaviour limits considerations of local context and dynamic processes in shaping climate change discussion and action post-event[21,22,29]. Yet we know from studies of contentious politics and social movements that such factors—specifically, threat, framing, political opportunity and resources—are critical for collective political action[30–33]. Drawing on concepts from the study of social movements, we conducted a fuzzy set Qualitative Comparative Analysis (fsQCA) of 15 communities impacted by an extreme weather event between 2012 and 2015 (Extended Data Fig. 1) to explore what combinations of event-, context- and response-related factors (Table 1) create climate change discussion and action post-event. We define climate change discussion and action (hereafter referred to as discussion) as a shared awareness of climate change as a social problem[11,34] and "nonroutine, collective, and public acts that involve claims on behalf of a larger collective"[35]. In operationalizing climate change discussion, we specifically focused on the links made between the event itself and climate change—in newspaper coverage of the event, by interviewees and in public statements by local opinion leaders (for example, a local or state elected official) and environmental organizations post-event—as an important first step in creating awareness and action about climate change. Our findings are based on case studies of these 15 communities, and include content analysis of 4,610 local newspaper articles and letters, 164 interviews with local residents and leaders, and secondary county-level data (on voting, community organizations, education levels and so on). Methods gives the details.

Post-event community climate change discussion

Among the 15 communities that experienced an extreme weather event, nine showed evidence of post-event discussion about the event's connection to climate change, albeit at low levels. In these cases, interviewees described post-event community discussion about the event's connection to climate change, local media wrote about the event's connection to climate change, and a local opinion leader invoked climate change in reference to the event or a local group responded to the event with activities intended to stimulate awareness of climate change. Yet, even in these communities, the number of local media articles about climate change was low (less than five for most), and no more than one opinion leader or local group publicly connected the event to climate change. Furthermore, none of those communities engaged in the kind of post-event mobilization activities frequently associated with social movements,

[1]Sociology / School of Public Policy, Oregon State University, Corvallis, OR, USA. [2]Public Health / Social and Behavioral Sciences, Oregon State University, Corvallis, OR, USA. [3]Civil and Environmental Engineering, Stanford University, Stanford, CA, USA. [4]Department of City and Metropolitan Planning, University of Utah, Salt Lake City, UT, USA. [5]Rural Sociology, The Pennsylvania State University, University Park, PA, USA. *e-mail: hilary.boudet@oregonstate.edu

Table 1 | Operationalization of outcome and causal conditions

Condition	Scoring		External source[a]	Theoretical expectation[b]
Outcome				
Community climate change discussion: Evidence of discussion and/or action related to climate change in connection to the event	0	% of interviewees: <30%; AND no. of articles: 0; AND no. of opinion leaders: 0; AND no. of local efforts, 0	NA	NA
	0.2	% of interviewees: <30%; AND no. of articles: 1; AND no. of opinion leaders: 0; AND no. of local efforts: 0		
	0.4	% of interviewees: 30–49%; AND no. of articles: 0; AND no. of opinion leaders: 0; AND no. of local efforts: 0		
	0.6	% of interviewees: 30–49%; AND no. of articles: 1; OR no. of opinion leaders: 1; OR no. of local efforts: 1		
	0.8	% of interviewees: >49%; AND no. of articles: 1; OR no. of opinion leaders: 1; OR no. of local efforts: 1		
	1	% of interviewees: >49%; AND no. of articles: >1; AND no. of opinion leaders: 1; AND no. of local efforts: 1		
Causal: event related				
Scientific attribution: Event with strong scientific support for climate change attribution and effect	0	Low scientific confidence attributing event type to climate change; includes extratropical cyclones, tornadoes and mudslides	National Academies[1]	+
	0.75	Moderate scientific confidence attributing event type to climate change; includes droughts (and associated wildfires), extreme rainfall (and associated flooding), tropical cyclones and extreme snow/ice		
	1	High scientific confidence attributing event type to climate change; includes extreme hot/cold events		
High-impact event: Event with high impacts on the community	0	Low fatality (<10 deaths) AND no Presidential Declaration	NA	±
	0.33	Medium fatality (≥10 deaths in >1 d) AND no Presidential Declaration		
	0.67	Medium fatality (≥10 deaths in >1 d) AND Presidential Declaration		
	1	High fatality (≥10 deaths in 1 d) AND Presidential Declaration		
Unusual event: Event type and/or timing unusual for area	0	Event type normal for area AND timing (time of day, season or duration) normal for the area	n/a	+
	0.75	Event type normal for area AND Timing (time of day, season or duration) NOT normal for area		
	1	Event type NOT normal for area		
Causal: context-related (pre-event)				
Democratic: Predominantly Democratic	0 to 1	Percentage of adults in county voting Democrat in the most recent presidential election results, calibrated as follows: fully out = 0.30, crossover = 0.50, fully in = 0.70	*New York Times* elections[52,53]	+
Education: High education levels	0 to 1	Percentage of adults 25+ years in county with bachelor's degree or higher, calibrated as follows: fully out = 0.18, crossover = 0.28, fully in = 0.38	US Census[54]	+
Environmental civic capacity: Local organizational capacity or media coverage of climate change issues	0 to 1	No. of articles in local newspaper referring to 'global warming' or 'climate change' in year prior to the event, calibrated as follows: fully out = 10, crossover = 75, fully in = 225 OR no. of environmental organizations per 1,000, calibrated as follows: fully out = 0.01, crossover = 0.05, fully in = 0.08	NA NCCS[55]	+
Causal: response-related				
Convergence frustration: Post-event challenges managing the convergence of volunteers, donations and/or media	0	No articles or interviewees indicated that the community experienced convergence challenges OR 1 article OR 1 interviewee indicated convergence management challenges	NA	-
	1	At least 1 article AND 1 interviewee indicated that the community experienced convergence management challenges		

[a]For conditions not scored using content analysis of newspapers or fieldwork. [b]For discussion and/or action. For the negation, theoretical expectations were reversed. NA, not applicable.

such as coordinated campaigns, testimony, protests, litigation or violence. The evidence suggests that in most communities, climate change was a marginal issue at best, especially compared with other post-event concerns, such as emergency response management and economic recovery.

Yet, there is a distinct qualitative difference between these nine communities and the remaining six. The remaining six communities showed virtually no post-event discussion of the event's connection to climate change, often with no interviewees describing post-event conversations about climate change and no articles mentioning climate change. Furthermore, none of them showed evidence of either local opinion leaders invoking climate change or local initiatives focused on climate change. In these cases, some interviewees admitted to avoiding engaging in discussions about

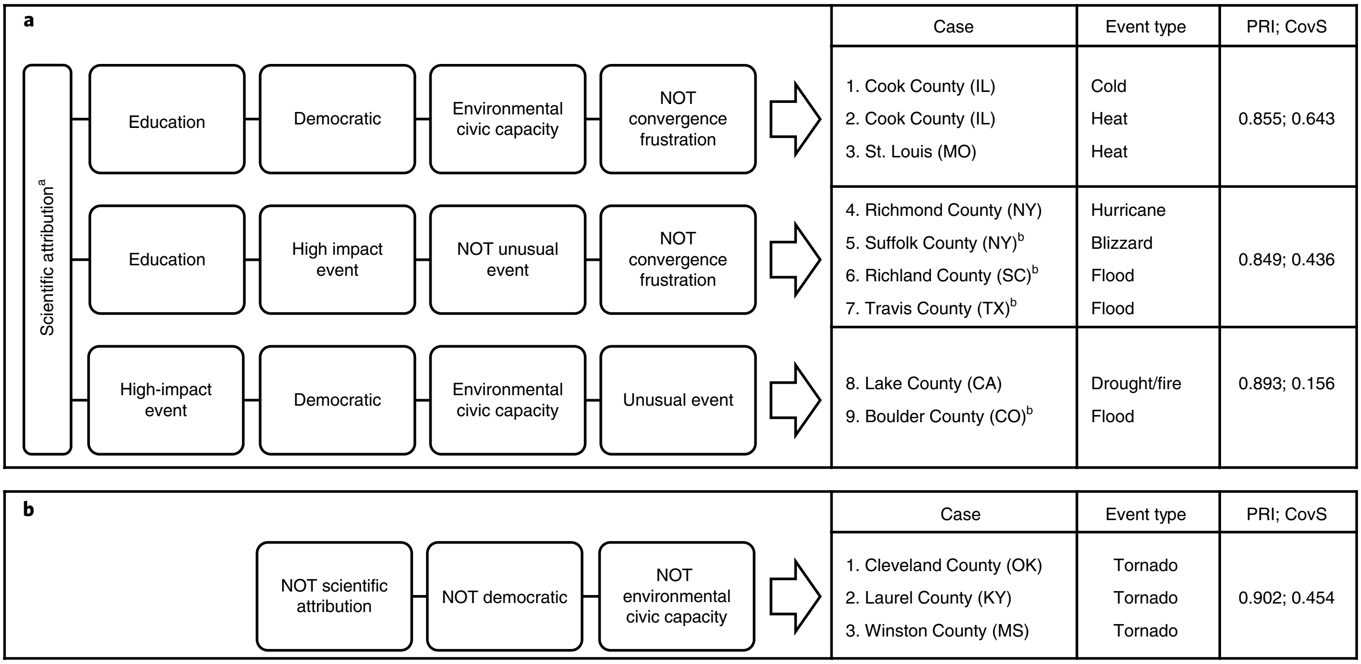

Fig. 1 | Causal recipes for the presence and absence of post-event climate change discussion. a, Causal recipes for the presence of post-event climate change discussion (solution test of sufficiency: consistency of sufficiency (InclS), 0.876; proportional reduction in inconsistency (PRI), 0.844; shared coverage score (CovS), 0.790.) **b**, Causal recipes for the absence of post-event climate change discussion. (Solution test of sufficiency: InclS, 0.920; PRI, 0.902; CovS, 0.454.) [a]Test of necessity: consistency of necessity, 0.878; relevance of necessity, 0.772; necessity coverage, 0.816. [b]Projects explained by more than one causal combination.

climate change at all due to its political nature; others audibly sighed or bristled with exasperation when asked about whether the event spurred discussions about climate change.

Factors associated with climate change discussion

Among the communities that exhibited post-event climate change discussion, the specific event was attributable to climate change with moderate-to-high confidence (for example, extreme heat and cold, drought-related wildfire). In these locations, the articles and interviewees tended to acknowledge the limits of attributing a single event to climate change, but nevertheless made explicit connections between climate change and the frequency and severity of extreme weather events. They pointed to the event they had experienced as an example of the types of weather events that could be expected in the future because of climate change and as a rationale for climate change action.

With scientific attribution as a necessary condition, our analysis yielded three distinct causal configurations, or 'recipes', that were sufficient to yield climate change discussion in the wake of an extreme weather event (Fig. 1a). Contextual factors were important in all the recipes, with the combination of Democratic political orientations, high pre-event environmental civic capacity (as it relates to climate change discussion) and/or high education levels setting the stage for discussion in all three recipes. Selected event characteristics also mattered, with unusual and/or high impact events contributing to climate change discussion in two recipes. Finally, the response experience, specifically the absence of challenges related to a post-event influx of media, volunteers and donations (referred to as 'convergence'[36]), made a difference in two recipes.

Exemplary cases of our first recipe, Democratic AND highly educated AND high environmental civic capacity AND low convergence frustration, included the Cook County (IL) and St Louis (MO) extreme heat events (Box 1), as well as the Cook County (IL) extreme cold weather event—events that are attributed to climate change with a high level of confidence by the scientific community. These urban areas, among the most heavily Democratic and highly educated cases in our sample, also experienced comparatively low impact events. Previous high-profile experiences (for example, the 1995 Chicago heat wave) had spurred considerable efforts to prepare for future extremes. These locations also hosted high levels of environmental civic capacity, with respect to the presence of environmental organization and/or pre-event media coverage of climate change. Moreover, these communities experienced low levels of convergence-related frustration. Indeed, none of the hot or cold events in our sample resulted in a large influx of media, volunteers or donations.

The second recipe, highly educated AND high impact AND not unusual AND low convergence frustration, explained four cases of post-event community climate change discussion: the Richmond County (NY) hurricane, the Suffolk County (NY) blizzard and the Richland County (SC) and Travis County (TX) floods (Box 2). These communities were highly educated with varied political orientations. All experienced a high-impact event that yielded a presidential disaster declaration and/or ten or more associated fatalities. Despite the high-impact nature of these events, they were not unusual in the affected area, and these communities experienced little to no challenges related to the convergence of media, donations or volunteers.

Our last recipe, Democratic AND high environmental civic capacity AND unusual event AND high impact, explained the two remaining cases of post-event climate change discussion—the Lake County (CA) fire (Box 3) and the Boulder County (CO) floods. Both communities experienced a high level of post-event discussion about climate change, which included strong opinion leadership and climate change activism. Indeed, the intensity of the Boulder County residents' beliefs about the event's connection to climate change overrode competing claims about scientific attribution[21]. Both counties experienced high-impact, unusual events. And, as for the first recipe, both communities are in Democratic-leaning counties with high pre-existing environmental civic capacity, although they are markedly different with respect to education levels.

Box 1 | St Louis (MO) heatwave

The St. Louis metropolitan area has a long history of heat events. Yet, the 2012 heatwave was noted by interviewees as being "unbearable" (Interview 109), with ten consecutive days of triple-digit temperatures[56] that resulted in at least 18 heat-related deaths. Among the cases in our sample, St Louis is the most liberal (84% Democratic). It is also home to multiple colleges and universities, which include the University of Missouri-St Louis and Washington University in St Louis and results in high education levels (42% bachelor's degree or higher). Prior to the event, the area had a moderate number of organizations per capita (0.05) and during the year before the event, local newspapers had published a moderate number of articles about climate change (82).

The city was well-prepared for the heatwave due to policies and initiatives put in place after an earlier heatwave in 1980 that caused more than 150 deaths. That event sparked a key collaborative initiative, Operation Weather Survival, and prompted the creation of an emergency response programme for future extreme heat events. Similarly, the non-profit Cool Down St Louis, which assists citizens with procuring air conditioners and keeping their power on, was created at the same time and continues to maintain an active presence in St Louis. The city's preparations and its regular exposure to less-intense heat-related events limited challenges related to either the initial emergency response or the convergence of volunteers, donation or media.

Newspaper articles and interviewees indicated that the community saw a strong connection between the 2012 St Louis heatwave and climate change, and often discussed this linkage in relation to other extreme events that had occurred throughout the Midwest. Although 2012 was extreme in temperature and heat-related deaths, 2011 was also a troubling time for the state of Missouri marked by six extreme weather events (one blizzard, two floods, two tornadoes and one heat wave/drought).

Local opinion leaders that we interviewed universally acknowledged a connection between the heatwave and climate change. For example, the Executive Director of the St Louis Audubon Society remarked, "There's no doubt in my mind that climate change … is part of it [the 2012 heatwave]" (Interview 106). Although there is no evidence of new organizations being formed in response to the 2012 heatwave, interviewees described ongoing activities by key conservation and environmental organizations to educate the public about climate change, which included activities by Bring Conservation Home, an urban land habitat restoration assistance and certification programme coordinated by the St Louis Audubon Society.

Despite this post-event climate change discussion, multiple interviewees also noted a significant divide between the more urban population in St Louis and more rural, agricultural areas in the west of the state. Yet, as one local climate scientist contended, even in the more conservative areas of the state, awareness is increasing. He went on to observe that, "the farmers, at least, know that something's up …" (Interview 108).

Factors associated with no climate change discussion

Our analysis yielded no necessary conditions for the absence of post-event community climate change discussion and only one combination of sufficient conditions: events not attributed to climate change that occurred in moderate-to-heavy Republican-leaning counties that lacked pre-existing environmental civic capacity (Fig. 1b). This recipe explained our three tornado cases: Laurel County (KY), Cleveland County (OK) (Box 4) and Winston County (MS). In these cases, few interviewees described post-event community discussion about climate change, and we found no engagement with the issue in news articles or by opinion leaders or local organizations.

The fsQCA process left unexplained the lack of post-event climate change discussion in three cases: the Yavapai County (AZ) fire, the Snohomish County (WA) mudslide and the Clark County (NV) extreme heat event. Each case represented unique combinations of conditions that yielded somewhat surprising outcomes. Two represented event types with high levels of scientific attribution: the Clark County (NV) extreme heat event and the Yavapai County (AZ) fire (where attribution is due to prefire drought conditions). However, residents appeared to view the events as being within the range of normal weather events and therefore not attributable to climate change. In Yavapai County (AZ), community members described the fire as a normal risk. Similarly, in Clark County (NV), triple-digit temperatures were described as normal by local media and interviewees, "It's a desert; it's hot here" (Interview 22).

In those cases, other factors were deemed responsible for the event. In Yavapai County (AZ), interviewees and local media coverage tended to attribute the fire's level of devastation to poor forest management and high fuel load, rather than climate change. Similarly, in Snohomish County (WA), interviewees attributed the mudslide to logging and land-use decisions. Moreover, the communities most proximate to the mudslide were politically and economically dissimilar from the surrounding county. While the case scored moderately Democratic based on county-level voting patterns, our site visit suggested a significant divide between the affected less-populated rural areas and more-populated urban areas within the county.

Discussion and conclusion

Although many of the extreme events we examined spurred significant mobilization for emergency response (volunteers, and donations for rescue and recovery efforts), we found they sparked little mobilization about climate change. In most of our cases, connections between the event and climate change remained rare, especially compared to discussions about the more immediate response and recovery concerns, both in interviews and post-event media coverage, even in places where climate change was discussed post-event.

Yet, there was a distinct qualitative difference between cases where community climate change discussion occurred and where it did not, which allowed us to uncover pathways to discussion and the lack thereof. Evidence of community discussion about the event's connection to climate change was found in a little over half of the cases, especially in communities primed to view these events as such (Democratic with pre-event media coverage of climate change) and with pre-existing resources to act (education and environmental civic capacity). Moreover, communities that engaged in post-event climate change discussion tended to have escaped challenges related to the convergence of volunteers, donations and media that can accompany these events. Our results thus suggest that a single extreme weather event may result in only limited local discussion about the event's connection to climate change, and then only in places where contextual factors (socio-demographics, resources and pre-event media framing) and event-related factors (scientific attribution, impact and response) facilitate such a connection. In other words, not all extreme weather events are created equal in terms of facilitating awareness and action about climate change[22]. As found in other studies of social movement activity around potentially

Box 2 | Travis County (TX) flooding event

In the early morning hours of 31 October 2013, flash floods associated with severe thunderstorms broke out in Travis County (TX), with the southeast Austin neighbourhood of Onion Creek hit the hardest. Emergency Medical Services reported that, "fast-moving flooding occurred when people were asleep, and more than 300 people called 911 to request help to get out of their rapidly flooding homes"[57]. Alongside the rapidly flooding Onion Creek, winds knocked down power lines, uprooted trees, damaged roads and resulted in four deaths[58]. In the aftermath of the flood, 12 homes were determined to be uninhabitable, whereas 531 homes were declared "too dangerous to live in"[59]. President Obama declared the Travis County 'Halloween Flood' a natural disaster in December 2013, a month-and-a-half after the event[60].

Travis County is largely Democratic (64%) and well-educated (46%), and exhibited a moderate per capita number of environmental organizations (0.06) and local articles about climate change (71). As multiple interviewees described, the county is also in 'Flash Flood Alley', an area in Central Texas prone to flooding due to a combination of intense rainfall events, steep topography and shallow soils[61]. As one Travis County Commissioner noted, "We've always had flash floods, we're always going to have them" (Interview 137). Interviewees also pointed out that heavy rains a few weeks earlier had left the soils saturated, which created conditions ripe for flooding. Moreover, the Onion Creek neighbourhood was built in a flood-plain, which leads to a persistent pattern of flooding, including this and three subsequent flooding events.

Travis County experienced a moderate level of post-event community discussion that associated the flooding event with climate change. Although some interviewees described the amount of rainfall as normal, most acknowledged the extreme nature of the event; a majority attributed the flooding to climate change. For example, another Travis County Commissioner said, "Climate change is making these extreme weather events even worse" (Interview 138). Likewise, the County's Emergency Management Coordinator stated, "I wouldn't be surprised if some of what we've seen could be attributed to climate change; we have seen some very strange weather patterns not just here, but all over the United States" (Interview 128).

However, this view was not universal. The Assistant Director of the Austin Disaster Relief Network connected the increased quantity and frequency of rain to El Niño, not climate change (Interview 124). Multiple interviewees, regardless of their views on climate change, tied the flooding event to land use and planning decisions. The local newspaper did not mention an explicit connection to climate change. And, although several individuals were noted for speaking out about climate change generally, there is no evidence that opinion leaders made explicit links between this flooding event and climate change. One interviewee described her rationale for not speaking out about that connection, despite a personal belief in it, as follows: "There wasn't any scientific data that I could point to, and … Because some of it just engenders a tremendous amount of pointless debate and argument, I am simply focusing on increasingly violent and deadly weather" (Interview 138).

At least one local non-profit organization, the Sierra Club in Austin, made the flood a central element of its climate change awareness and advocacy campaign, which included working with the city of Austin to develop a long-range energy plan. An organizer for the Sierra Club in Austin stated:

> The flooding, the fires and the drought were our main motivating points. It was what our volunteers and members talked about every other word. It was a central piece. We had some folks from the flood impacted zones speak and testify and share their stories; we took stories from them and put them on our Facebook page. (Interview 136)

The community experienced major frustrations with initial emergency response efforts, including the failure of two flood gauges on Onion Creek, delayed notification and inadequate emergency response services. There was also significant angst about longer-term clean-up and recovery efforts, which included delays in the Federal Emergency Management Agency funding award process. That said, we found no evidence of challenges related to the management of donations, volunteers or media. In fact, after an initial gathering of 400 volunteers, the Executive Director of the Austin Disaster Relief Network asked, "Where is the rest of Austin? … we are going to need thousands"[62].

contentious issues[37–40], no one factor in isolation is enough to generate collective action; it is, instead, their combination that matters for mobilization.

Indeed, recent scholarship has questioned the effectiveness of 'proximizing' climate change (for example, by highlighting extreme weather experience) to motivate individual awareness and action[41–43]. These scholars argue that bringing climate change closer to home will not necessarily result in increased action unless such events are interpreted as signals of climate change, affect places and things one cares about, and possible responses are deemed reasonable, achievable and efficacious[41–44]. We provide evidence that similar factors matter at a community level, in particular, contextual factors that facilitate or impede the interpretation of an event as a climate signal.

In some of the higher-impact events examined here, interviewees suggested that the search, rescue and recovery process associated with the event appeared to be more than enough for residents to handle, without adding the spectre of climate change to the mix. More proximate concerns—emergency management processes, the organization of the large influx of volunteers, handling donations and media attention, local rebuilding and economic redevelopment efforts and dealing with long-term psychological impacts on residents—often dominated interviews with local leaders. Some interviewees cautioned that even broaching the topic of climate change amid a disaster recovery could be perceived as using a community's loss to further a political agenda. From an environmental activist in Boulder County, "There's a balance [in] how to address climate change without looking like you're trying to capitalize on the event … When people are suffering, when they've just been through a traumatic event—people died, you know—you have to be very sensitive and careful not to just look blasé about, 'Oh, look, climate change again'" (Interview 7). A local newspaper editor in Snohomish County, Washington, was even blunter, "A friend … wrote me right away and said the experts had been talking about climate change as the cause of this …. When you're in the middle of it, that's not the time … I mean, we haven't even decided what we're going to call the event itself … how are we going to have a discussion about climate change?" (Interview 99). Similarly, from an interviewee in Richland County, "We were dealing with people that lost their homes … or had their businesses washed away. The whole discussion about climate change was pretty far off from people's immediate concerns … People were more focused on the immediate issues, like, let's get rid of this crumbling building or let's get our water turned back on, more than anything else" (Interview 79).

Shifts in US opinions on climate change in recent years[45] suggest that climate change may become a more acceptable topic of

Box 3 | Lake County (CA)

The 'Valley Fire' that occurred in Lake County, California, in autumn 2015 was the third most destructive wildfire in California's history at the time[63]. Governor Brown declared a state of emergency on the day of the fire, and President Obama declared it a major disaster a week later[64,65]. The fire occurred during a major drought in California that began in 2012 and lasted until 2017. It burned more than 75,000 acres, caused the evacuation of nearly 20,000 residents (roughly a third of Lake County's population) and destroyed 1,958 structures[63,66–69]. Ultimately, the fire displaced 3,700 residents, nearly 6% of Lake County's population. Prior to the event, Lake County was majority Democrat (56%) with relatively low levels of education (16%), but with the highest per capita environmental organizations (0.16) in our sample and a considerable number of local articles on climate change (91).

Most interviewees and multiple articles described widespread discussion about the event's connection to climate change, even while stating that climate change did not directly cause the fire. They also described a likely link between climate change and pre-existing drought conditions, which included warmer temperatures, reduced snowpack and bark beetle infestations, all of which raised the risk, speed and size of the fire. Several interviewees indicated that the event encouraged discussion about climate change, even though they personally continued to question the anthropogenic nature of climate change. For example, in our interview with the California Fire Division Chief and Fire Chief for South Lake County Fire Protection District, he connected the intensity and rapid spread of the fire to climate change, though not necessarily human caused:

I don't know how much of it is human caused. I'm not a scientist. I don't have an agenda. I just look at the way fires have been responding in the last several years and it is different. The colors of the fire are different and I'm not the only one saying that; there's certain colors, a deep red, that we used to not see …. Things just happen quicker. (Interview 54)

The climate change message was heavily bolstered by California Governor Brown's discussion of the Valley Fire and his strong climate change advocacy. Staff at the local Community Action Agency noted the Governor's role as an opinion leader guiding California's conversation about the connection between climate change and wildfires (Interviews 60 and 61).

Nevertheless, not everyone agreed on the connection between climate change and the wildfire. For example, one interview participant described the fire as "a result of 70 years of fire suppression and a weather event" (Interview 63). Similarly, a Lake County Supervisor said that, although others in the community connected the drought to climate change, he did not believe that climate change caused the fire or drought. "[The] climate's been changing ever since Earth formed. So, it does happen, but it's not as big a thing for me" (Interview 62).

Despite a relatively effective emergency response effort and a highly collaborative community response, Lake County experienced several post-event convergence issues related to donations, which included an estimated 500,000 British pounds of donations that required the rental of a warehouse for storage. The affected communities also experienced significant challenges related to longer-term recovery efforts, specifically delays and bottlenecks associated with Federal Emergency Management Agency procedures and local government and permitting, with one interviewee lamenting, "rebuilding has been a disaster in several places" (Interview 59).

conversation post-event. However, a recent *New York Times* article concluded that Midwestern mayors in areas that flooded this spring deemed climate change a topic best avoided due to its political nature[46]. And, although such shifts in public opinion may, in part, be driven by extreme weather experience, particularly among conservatives[47], our results indicate that it may take time for such shifts in opinion to influence public discussion, particularly in Republican-leaning communities. Future research should examine in more detail the prevalence of climate change discussion post-event compared to other topics and relationships to policy action for these and more recent events to better understand where, how and when to discuss climate change in the wake of extreme weather events. Previous research at the national level has found policy change post-event to be rare[48,49], but it could be more common at the local level.

It is worth noting that timing could be important here. Our newspaper analysis ran one year post-event; our interviews were conducted, on average, three years post-event. Our results may have been different using different timescales between events, media coverage and interviews. Yet, our findings do suggest that discussions about climate change might be particularly fruitful around events scientists more readily attribute to climate change and where contextual factors like Democratic political leanings and pre-event media coverage can facilitate discussion, and where resources like environmental organizations can facilitate action. Other factors that we do not explore here, such as public and stakeholder engagement processes post-event[50] and the political leanings of local newspapers[51], could also prove important in facilitating or impeding action and should be explored in future research.

We did find some evidence of communities making connections between extreme weather events and climate change, with the event often discussed as occurring in quick succession with other extreme events. For example, in St Louis, participants hypothesized that it was not a single event but successive extreme events that occurred throughout Missouri—particularly in 2011, when the state experienced a blizzard, multiple floods, multiple tornadoes and extreme heat and drought—which prompted residents to consider the impacts of climate change (Interviews 106 and 108). Such findings suggest that, as more people personally and virtually experience the impacts of multiple climate-related extreme weather events, more discussion and possibly mobilization may occur if such events are attributed to climate change. Again, future research could test the role of cumulative extreme event experience in shaping views and actions on climate change.

Our work challenges the notion that a single extreme weather event will yield rapid social mobilization around climate change. Yet, this finding does not mean communities may not make important changes post-event to ensure more effective responses to future events (Box 4). Future research should examine local policy action post-event as adaptive actions that may occur without discussions or activism about climate change, particularly in communities where the local contextual factors make climate change discussion difficult.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41558-019-0641-3.

Box 4 | Cleveland County (OK)

On the afternoon of 20 May 2013, an EF5 tornado struck the community of Moore, Oklahoma, a suburb of approximately 60,000 people located 10 miles south of Oklahoma City. The tornado, which travelled 14 miles and was 1.1 miles wide at its widest[70], injured 350 people and left 24 dead, including 9 children[71,72]. The Oklahoma Department of Emergency Management estimated that 1,150 homes were destroyed[73], in addition to multiple public and private facilities. President Obama toured the area immediately after the storm and declared a major disaster. In total, damages to commercial and residential property were estimated at US$2 billion[74]. At the time of the event, Cleveland County was predominantly Republican (62%) and was moderately well-educated (31% bachelor's degree or higher), but had a low environmental civic capacity, with an environmental organization presence (0.03) and only three articles on climate change published in local media.

The tornado was not the community's first experience with catastrophic storms. An EF5 tornado hit the community in 1999, followed by an EF4 tornado in 2003[75], both of which destroyed over 400 homes (Interviews 31 and 35). Even though some interviewees stated that the 1999 storm had been considered a "once in a life time occurrence", they indicated that the city learned from those experiences by taking such steps as establishing storm shelters, re-evaluating insurance coverage and contracting in advance for debris removal. The city was thus better prepared to address the 2013 storm and its aftermath. Indeed, citizens were for the most part satisfied by the local government's response and recovery efforts, even though local officials described challenges with volunteer-, donation- and press-related convergence issues. Yet, the local Economic Development Director stated that the 2013 storm was "monumentally worse and different" than previous storms, in part because of its timing and the emotional toll due to the loss of children's lives (Interview 37). Litigation related to the deaths of the children killed at an elementary school was still ongoing at the time of our interviews.

Despite the high impact and unusual timing of the tornado, not one newspaper article mentioned climate change when reporting about the event. Moreover, no interviewees discussed the connection between the tornado and climate change without prompting. When asked if climate change was discussed in the community in relation to the storm, interviewees said it was not. Little credence was given to debates in the national media about the possibility that climate change played a role in exacerbating the storm[76]. Even a representative of the local Sierra Club chapter rejected the view that climate change may have contributed to the event's occurrence or severity. Instead, most interviewees were resolute that tornadoes are simply a risk that comes with living in Oklahoma. As the Emergency Management Director stated:

> There have been tornadoes in Oklahoma since the beginning of time. There will be tornadoes in Oklahoma long after I'm gone … Doesn't matter where you live, there's always something. And severe storms happen to be one of ours. We plan for them. We build for them. (Interview 35)

The community mobilized extensively in its response and recovery effort. This resulted in several initiatives, which included the formation of an advocacy group, Shelter Oklahoma Schools, and the passage of a school bond measure to provide funding for storm shelters in Moore Public Schools. As might be expected, given the low level of post-event community discussion about climate change, there was no activism by local organizations around climate change, and no evidence that opinion leaders actively linked the event to climate change.

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Methods

Case selection. The Spatial Hazard Events and Losses Database for the United States (SHELDUS)[77] provided the main sampling frame for the study. This database provides county-level hazard data—specifically, event timing, location, property losses, crop losses, fatalities and injuries—for US natural disasters. We limited our search to counties with weather events that caused fatalities and/or injuries between 2012 and 2014 ($N=392$). These data were supplemented by 2015 weather events reported by National Oceanic and Atmospheric Administration's Billion-Dollar Climate and Weather Disasters[78], which were not yet available in the SHELDUS database when we selected our cases. To avoid oversampling wealthy and/or agricultural communities and undersampling hot and cold events, which create little property damage, we did not use property or crop damage as selection criteria[79].

As we were interested in 'extreme' weather events, we selected events that resulted in at least four countywide fatalities ($N=39$), although we learned later from newspaper articles that one case (Cook County heat) resulted in only one countywide fatality. Among those cases, we further limited the criteria for inclusion based on accessibility of a local (municipal or county) newspaper of record from the date of the event to 12 months after the event, with coverage of the identified event in sufficient detail (50 or more articles) ($N=24$). From the resulting list of 24 potential cases, we then selected cases to cover all weather hazards (flooding, tornado, wildfire, landslide, winter weather, heat and hurricane or tropical storm) and all regions of the United States (northeast, southwest, west, southeast and Midwest). For multiple, colocated counties that experience the same event as part of the same weather system, in our sample we typically included the most impacted county in terms of fatalities. Additionally, if the same weather event occurred in the same county two years in a row, only one of these years was selected as a case—again, typically the event with the highest number of fatalities. When weighing the inclusion of competing cases, a diversity of hazard types and geographical areas sometimes superseded the number of fatalities from the event (Extended Data Fig 1 and Supplementary Table 1).

Data collection. Following the approach taken by McAdam and Boudet[34] for advanced preparatory fieldwork, we first systematically searched the relevant local newspaper for articles and letters to the editor on the weather event in the year after its occurrence using the terms provided in Supplementary Table 2. We supplemented this search with a systematic search for the terms 'climate change' and 'global warming' in the year prior to the event and within the articles and/or letters focused on the event in the year after. All the relevant newspaper articles ($n=4,610$) and editorials ($n=203$) were then coded to identify leaders and important actors, groups and organizations, events and meetings, and proposed policies. This coding process was developed iteratively by five members of the research team using the Moore County (OK) tornado and Snohomish County (WA) landslide cases. H.W. completed this coding process for 12 of the 15 cases; the other three cases were coded by two other members of the research team, in close collaboration with H.B. and H.W. On completion of the coding process, coders wrote a memo (Supplementary Note 1 gives the template)—essentially the beginnings of a case narrative, an important component of fsQCA—that outlined learnings from the case, which was reviewed and edited by H.B. for consistency across cases and shared with the researcher who would conduct the case's fieldwork and interviews. We also collected and shared relevant socio-demographic, environmental civic capacity and event-related data available from secondary data sources with the researcher who would conduct the case's fieldwork and interviews, which included data from the US Census Bureau's American Community Survey[54], the National Center for Charitable Statistics[55], Yale Climate Opinion Maps[80], *New York Times* election website[52,53] and the National Academies Report on Event Attribution[1].

The researcher who conducted the case's fieldwork and interviews then used the list of leaders, important actors and organizations generated from newspaper coding to select and contact potential interview participants, with the goal to reach a panel of informants for each case that included at least one local elected official, county or city staff member, newspaper reporter, leader of a local community organization and religious leader[81]. If an environmental organization was not mentioned in the newspapers, we explicitly sought interviews with a representative from at least one local environmental organization, using online searches or snowball sampling, as the most likely type of organization to facilitate climate change discussion. Interviewees were invited to participate in a study about community reactions to extreme weather, with no specific mention of climate change or global warming. The field researcher also observed any relevant public meetings that took place during these visits and, when possible, conducted additional interviews as suggested by our initial participants. Five members of the research team completed the fieldwork for the project in close consultation with H.B.; fieldwork for five of the 15 cases was completed by H.S.

Over the course of two years from October 2015 to October 2017, we conducted 164 interviews, logging a total of 128 h. The average interview lasted 47 min, ranging from over 3 h to less than 10 min (Supplementary Table 3 gives a summary of interviews by case and Supplementary Table 4 gives additional details on interview data). Interviews were semistructured in nature; participants were asked about the community prior to the event, the event's major impacts generally and (if not mentioned) specifically about the event's impacts on climate change discussion, awareness and/or action (Supplementary Notes 2 and 3 give interview protocols). Interviews were recorded with participant permission; researcher notes taken during the interview itself were supplemented using these recordings. In addition, H.W. reviewed all the recordings and transcribed all the content in which climate change or global warming was mentioned or discussed. For each case, the researcher who conducted the fieldwork and interviews prepared a detailed research memo (Supplementary Note 4 gives template) building from the newspaper coding memo and consolidating information from interviews, newspaper articles and secondary sources—which again was reviewed and edited by H.B. for consistency across cases. We used these memos to operationalize our causal and outcome conditions. The length of time between the event and fieldwork varied from 19 to 60 months, with an average of 35 months. The study was approved the Oregon State University Institutional Review Board (no. 6303); informed consent was obtained from all the participants.

fsQCA. We used Ragin's[82,83] method of fsQCA to identify the combination of conditions that yield varying levels of emergent collective discussion related to climate change in the selected communities. fsQCA is a set-theoretic approach that uses a combination of qualitative and quantitative data to identify necessary and sufficient combinations of causal conditions.

There are several distinct advantages of fsQCA, especially in comparison with traditional statistical approaches. First, the method accommodates conjunctural causation, or the possibility that multiple causes act in combination to influence an outcome. Second, fsQCA permits equifinality, or the potential for multiple paths to lead to a similar outcome. Third, the approach explicitly allows for asymmetric outcomes, or the possibility that the positive expression of an outcome (for example, climate change discussion) is influenced by different factors from the negative expression of the outcome (for example, no climate change discussion). Finally, in contrast to traditional qualitative comparative analysis, which only permits the use of binary (0/1) condition and outcome scores, fsQCA accommodates scoring along a 0–1 scale, and thus permits more nuanced scoring and calibration of causal conditions and outcomes[82–85].

In standard fsQCA methods, a causal condition (or combination of conditions) is necessary when the outcome can be shown to be a subset of the condition. In other words, for all (or almost all) cases in which the condition is present, the outcome is also present, and there are no (or few) examples of the outcome being present in the absence of the condition. A causal condition is sufficient when it can be shown to be a subset of the outcome. That is, when the sufficient condition is present, the outcome is also present and there are no (or few) examples of the outcome being absent in the presence of the condition[84,86].

Standard measures to assess the strength of these causal relationships include consistency and coverage. Set-theoretic consistency reflects the degree to which cases with the same condition or combination of conditions also share the outcome of interest[87]. The standard statistic for assessing necessity, consistency of necessity, ranges from 0 to 1, and a consistency of necessity score above 0.9 provides strong evidence of necessity. The standard statistic for assessing sufficiency, the PRI, also ranges from 0 to 1, and a PRI score above 0.8 provides strong evidence of necessity[88]. Set-theoretic coverage reflects the degree to which a specific condition or combination of conditions explains occurrences of the outcome of interest. Coverage is typically measured by raw CovS; a higher CovS typically indicates that the given causal condition is more empirically relevant[87]. Supplementary Note 5 gives further detail on our rationale for selecting fsQCA and a more formal explanation of these measures.

Following Ragin[83], our analysis began with testing the necessity of all hypothesized causal conditions in Table 1. We then tested the sufficiency of our remaining conditions, grouped according to theoretical considerations, qualitative case data and assessments of consistency and coverage; the intermediate solution is presented in the manuscript results. We analysed the causal configurations that led to the outcome and the negation of the outcome separately. For both tests of necessity and sufficiency, we assessed our results based on standard measures of consistency and coverage. Finally, we conducted substantial sensitivity testing to assess the robustness of our results to a variety of changes to the calibration and combinations of conditions, following the exhaustive enumeration approach recommended by Schneider and Wagemann[84]. All the fsQCA analyses were conducted in R 3.5.2 using QCA package 3.4 described by Thiem and Duşa[89] and Duşa[90].

Supplementary Note 5 gives detailed information about the analytical process and results, which includes operationalization (Supplementary Tables 5–12), calibration assumptions (Supplementary Table 13), tests of necessity (Supplementary Tables 14 and 15 and Supplementary Figure 1), directional expectations (Supplementary Table 16), truth tables (Supplementary Tables 17 and 18), solutions from tests of sufficiency (Supplementary Tables 19 and 20) and sensitivity tests (Supplementary Tables 21 and 22).

Operationalization. We operationalized our outcome and causal conditions based on both theoretical considerations and our empirical evidence derived from the cases. Our primary outcome of interest represents the degree to which the community engaged in post-event discussion and action that linked the event to climate change. We scored community discussion by searching newspaper articles and interview transcripts for evidence of post-event community discussion about climate change. We included four indicators, informed by our fieldwork, to measure community discussion: the proportion of interviewees who described the occurrence of community discussion about the event's connection to climate change, the number of articles in the local newspaper that discussed the event's connection to climate change in an affirming or neutral manner, the number of opinion leaders who publicly spoke out about the event's connection to climate change and the number of local efforts to stimulate awareness about climate change in connection to the extreme weather event. Table 1 and Supplementary Tables 5–13 provide detailed information about our scoring process for our outcome and causal conditions, described below.

Based on findings from the literature, we focused on three types of causal conditions, aimed at representing both socio-demographic and physical contextual factors that might influence climate change discussion and action post-event, including event-, context- and response-related characteristics.

We considered three major types of event characteristics as having the potential to influence post-event community discussion about climate change: (1) scientific attribution (that is, expert determination of the influence of climate change on specific events), (2) impact and (3) the degree to which an event was unusual for the area. Our cases included six distinct types of extreme weather event: extreme heat and cold ($n=5$), wildfire ($n=2$), tropical storm (that is, hurricane) ($n=1$), severe convection storm/tornado ($n=3$), flood ($n=3$) and mudslide ($n=1$).

According to the National Academies report[1], the scientific community has the highest confidence in attributing severe heat and cold events to climate change, followed by droughts, extreme rainfall, tropical cyclones and extreme snow and/or ice. Flooding is difficult to attribute to climate change because a variety of factors, which include human decisions around land use and infrastructure, come into play. However, as the three cases of flooding in our sample were preceded by extreme rainfall, which is attributable to climate change with a moderate level of confidence, we scored each of these cases as 0.75. Wildfires are among the types of weather events that the scientific community has the lowest confidence in event attribution. However, as the wildfires in our sample were associated with droughts and the scientific community has moderate confidence that droughts are among the events that can be attributed to climate change, we scored these cases higher. The occurrence of severe convection storms, which includes tornadoes, are events for which the scientific community has the lowest level of confidence in attributing to climate change. And, although landslides may be affected by climate change, the National Academies report[1] does not explicitly address the attribution of mudslides. Heavy rainfall preceded the mudslide case in our sample, but the connection was not strong enough to suggest scientific attribution.

We also expected that the event's impact—in terms of fatalities, injuries, property damage and perceived harm—would affect the likelihood of a community to engage in climate change discussion. We posited that higher impact events were more likely to act as focusing events to draw attention to potential problems related to the event[91]. However, we also recognized that communities recovering from the highest impact events might be unable to consider or address issues beyond disaster relief. In contrast, communities with more moderate impacts might be able to retain the emotional and physical capacity to engage in climate change awareness discussion. We combined three indicators to score event impact: presidential major disaster declaration, the number of deaths that resulted from the event and the time frame during which deaths occurred. The highest-impact events were marked by a major disaster declaration and ten or more fatalities that occurred over the course of a single day.

In considering an event's unusualness, we posited that communities that had experienced and recovered from similar weather events would be less likely to link subsequent events to climate change, especially if such events were viewed as routine. We used information about the degree to which the weather event was considered unusual, based both on the type and timing (time of day, season or duration) of the event. The degree of normalcy for the event was based both on newspaper articles and interviewee responses.

We operationalized several contextual causes, which included political orientation, level of higher education attainment and environmental civic capacity, all of which were measured prior to the weather event. Based on considerable research that indicates Democrats are significantly more likely to believe in human-caused climate change than Republicans[8,12,92–95], we expected political orientation to be an important factor in shaping post-event climate change discussion. Similarly, we hypothesized that highly educated communities would be more likely to engage in discussion about climate change, based on the strong interaction between political orientation and education[8,96–98]. For each county in which our 15 events took place, we used extant data to measure the percentage of voters who voted Democratic in the 2008 Presidential election[61] and the percentage of adults 25 years and older who had earned a bachelor's degree. These data were calibrated using Ragin's[88] direct method of calibration for conventional interval-scale indicators.

Finally, considerable research has demonstrated that both organizational presence[24,30,31] and media framing[92,99,100] are important factors in shaping public perceptions and action around controversial issues. We expected that communities with existing capacity for environmental mobilization and/or prior local media coverage of climate change would be more likely to engage in post-event climate change discussion. We measured and calibrated these concepts separately. To measure the local capacity for environmental mobilization, we used data from the National Center for Charitable Statistics to calculate the number of environmental organizations in each county prior to the event. To assess local media coverage of climate change, we conducted a systematic search for local newspaper articles that referred to 'climate change' or 'global warming' in the year prior to the event. The two measures were used to form a macrocondition, which we refer to as environmental civic capacity. The use of the number of non-profits per capita as a proxy for organizational capacity and newspaper data to assess framing is not uncommon in the literature on social movements or environmental sociology[39,101,102].

Unlike the previous conditions, which were identified a priori as we designed our research study, one of our causal conditions—difficulty managing the convergence of low-priority material donations, spontaneous volunteers and media coverage—emerged from our case study work. The identification of unexpected or emergent phenomenon is a strength of qualitative research. The literature on disaster recovery suggests that convergence is an important but understudied logistical challenge to response and recovery efforts[37,103,104]. Yet, this issue has yet to be considered in studies about the relationship between extreme weather experience and climate change views and actions. We observed that, in some cases, difficulties in managing material, volunteer and media convergence also impeded a community's capacity and willingness to engage in discussion and action related to climate change.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Data used in this analysis are available at: https://osf.io/kxm23/.

Code availability

Programming code used in this analysis is available at: https://osf.io/kxm23/.

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Author contributions

H.B. conceived the study, secured funding, supervised the project and wrote the paper. L.G. analysed data and wrote the paper. C.Z., H.S. and H.W. collected data and contributed to the analysis.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41558-019-0641-3.

Supplementary information is available for this paper at https://doi.org/10.1038/s41558-019-0641-3.

Correspondence and requests for materials should be addressed to H.B.

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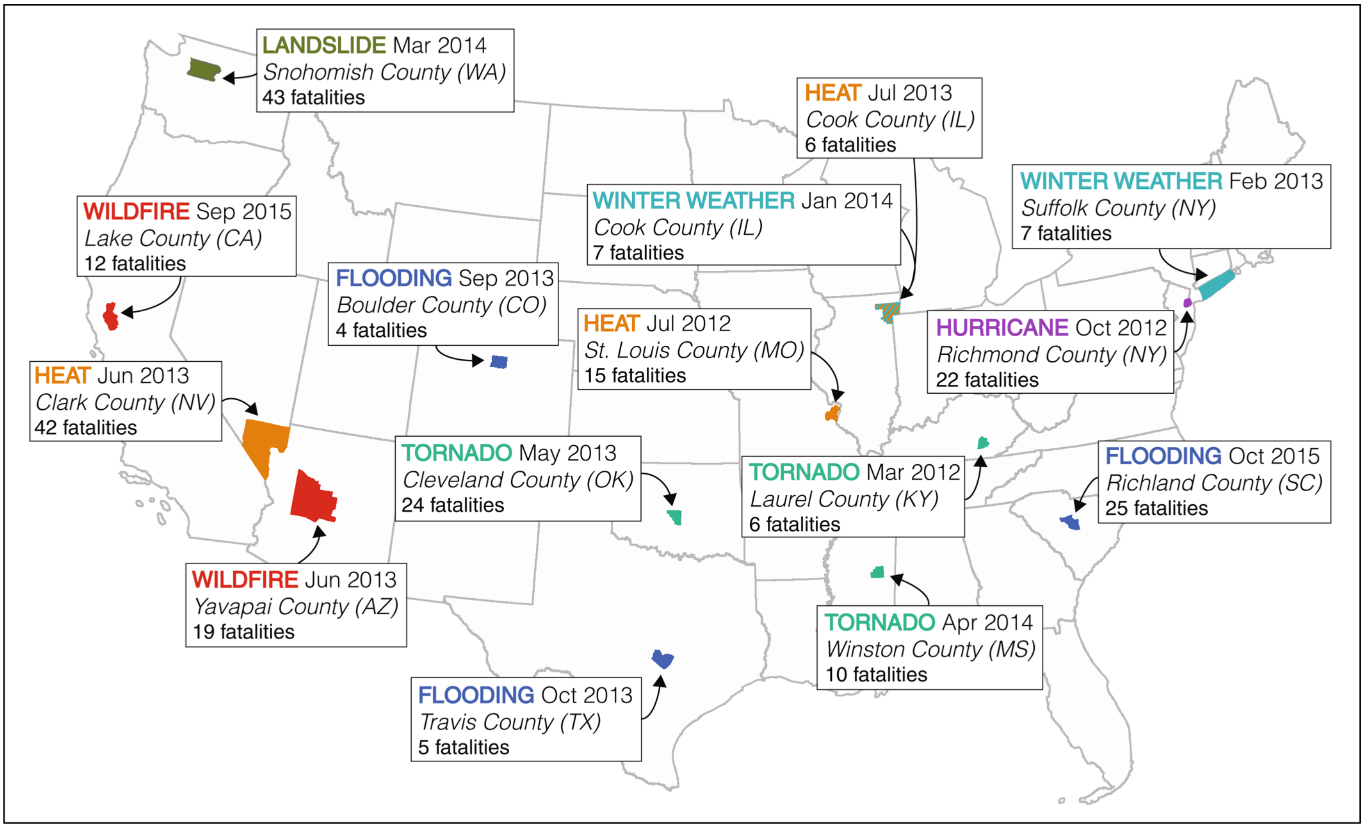

Extended Data Fig. 1 | Map of selected cases of extreme weather events, 2012–2015.

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☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used.
Data analysis	All fsQCA analyses were conducted in R 3.5.2 using the 'QCA' package 3.4.

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Behavioural & social sciences study design

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Study description	Mixed methods (fuzzy set/Qualitative Comparative Analysis)
Research sample	The Spatial Hazard Events and Losses Database for the United States (SHELDUS) provided the main sampling frame for the study. This database provides county-level hazard data – including timing, location, property losses, crop losses, fatalities and injuries – for U.S. natural disasters.
Sampling strategy	We limited our search to counties with weather events that caused fatalities and/or injuries between 2012 and 2014 (N=392). These data were supplemented by 2015 weather events reported by NOAA's Billion-Dollar Climate and Weather Disasters, which were not yet available in the SHELDUS database when we selected our cases. We did not use property or crop damage as a selection criterion to avoid over-sampling wealthy communities and under-sampling heatwaves, which create little property damage. After reviewing the initial sample, we selected events that resulted in at least four countywide fatalities (N=39) to strike a balance between community impacts and the likelihood of media coverage. Among those cases, we further limited the criteria for inclusion based on accessibility of a local (municipal or county) newspaper of record from the date of the event to 12 months after the event, with coverage of the identified event in sufficient detail (50 or more articles) (N=24). From the resulting list of 24 potential cases, we then selected cases to cover all weather hazards (flooding, tornado, wildfire, landslide, winter weather, heat, and hurricane/tropical storm) and all regions (Northeast, Southwest, West, Southeast, and Midwest). For multiple, co-located counties experiencing the same event as part of the same weather system, we typically included the most impacted county in terms of fatalities in our sample. Additionally, if the same weather event occurred in the same county two years in a row, only one of these years was selected as a case – again typically the event with the highest number of fatalities. When weighing inclusion of competing cases, a diversity of hazard-types and geographic areas sometimes superseded the number of fatalities from the event.
Data collection	We first systematically searched the relevant local newspaper for articles and editorials on the weather event in the year after its occurrence. We supplemented this search with a systematic search for the terms "climate change" and "global warming" in the year prior to the event and within the articles/letters focused on the event in the year after. All relevant newspaper articles (n=4,610) and editorials (n=203) were coded to identify leaders and important actors, groups and organizations, events and meetings, and proposed policies. We also collected relevant socio-demographic, organizational capacity and event-related data available from secondary data sources, including the U.S. Census Bureau's American Community Survey, the National Center for Charitable Statistics, Yale Climate Opinion Maps, National Academies Report on Event Attribution. We used newspaper and secondary data to both develop a narrative of the case and identify potential interview participants for a site visit. Finally, we visited each site to conduct interviews with key participants identified in the newspapers, including local and state elected officials, county/city staff, newspaper reporters, leaders of local community organizations, and religious leaders. We conducted 164 interviews, logging a total of 128 hours. The average interview duration lasted 47 minutes, although the interview duration ranged widely, from over three hours to less than 10 minutes. We also observed any relevant public meetings that took place during these visits and conducted additional interviews as suggested by our initial participants when possible. We then used these memos to operationalize our causal and outcome conditions.
Timing	2014 to 2019
Data exclusions	n/a
Non-participation	Response rates for interview requests varied by case with a minimum of 27% and a maximum of 85%. Average response rate across cases was 55%.
Randomization	n/a

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Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Human research participants

Policy information about studies involving human research participants

Population characteristics	n/a - Interviewees were not selected on the basis of gender, age or other factors but instead as a "panel of informants" based on the role(s) they played in disaster recovery efforts.
Recruitment	Participants were identified via media coverage of the event and recruited via email and/or phone. We also included at least one representative from a local environmental group, even if not mentioned in media coverage. We used snowball sampling to supplement our list of potential interviewees from media coverage.
Ethics oversight	The study was approved the Oregon State University Institutional Review Board (#6303); informed consent was obtained from all participants.

Note that full information on the approval of the study protocol must also be provided in the manuscript.